

DESKTOP APPLICATION

TODDLER

Interactive Learning Application for Children

01 • LEARNING CONTENT

Natural World Info

Interactive learning content covering fruits, vegetables, animals, and birds tailored for children.

02 • ARCHITECTURE

VB.NET & MS Access

Built as a Windows desktop application integrated with a Microsoft Access database.

03 • DATA CONTROL

CRUD Management

Complete administration capabilities allowing users to Add New, Save, Update, and Delete entries.

Purpose

A simple learning application aimed at students below 5th grade, designed to make basic natural-world information easier and faster to learn.

Learning Areas

Fruits, vegetables, animals and birds are presented with item details and pictures.

Management

An Edit area supports category selection and database operations: Add New, Save, Update and Delete.



Desktop Application

User-friendly desktop application built for smooth classroom interactions.



Digital Learning

Reduces dependence on manual charts and physical notebooks.



Data Storage

Stores item information securely for easy later retrieval.



Dual Utility

Supports both interactive learning and seamless data maintenance.

Existing Approach

The report describes a largely manual process relying on physical tools and high effort.

Manual Process

Uses physical charts, boards, notebooks, and heavy teacher/parent effort.

Time-Consuming & Repetitive

Inefficient workflows and repetitive tasks for teaching.

Vulnerable to Data Loss

Physical documentation carries high risk of loss or damage.

Objectives of TODDLER

Computerize learning to create an accessible, streamlined educational system.

Computerized Information

Makes learning content quickly accessible with improved storage and retrieval.

Reduce Manual Effort

Minimizes repetitive teaching tasks and parent/teacher workload.

Interactive Learning

Provides a modern, engaging, and interactive digital way for children to learn.



Language

VB.NET



IDE

Visual Studio 2022



Database

Microsoft Access



OS

Windows 11



Processor

AMD PRO A4-4350B



RAM

4 GB



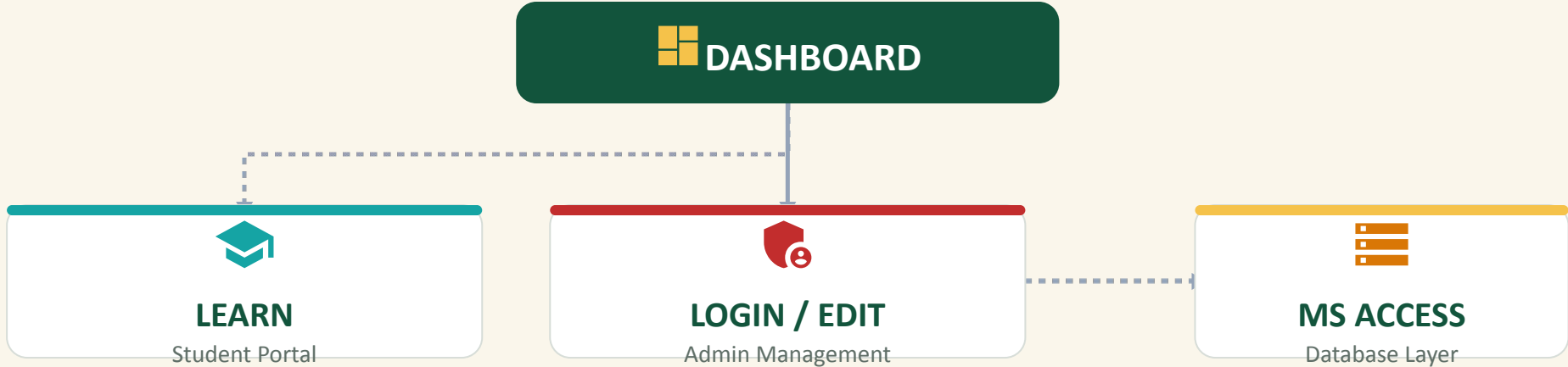
Hard Disk

500 GB



Monitor

14-inch LCD



Learn Module Categories

Interactive learning topics accessible directly through the Learn module:

 **Fruit**

 **Vegetable**

 **Animal**

 **Birds**

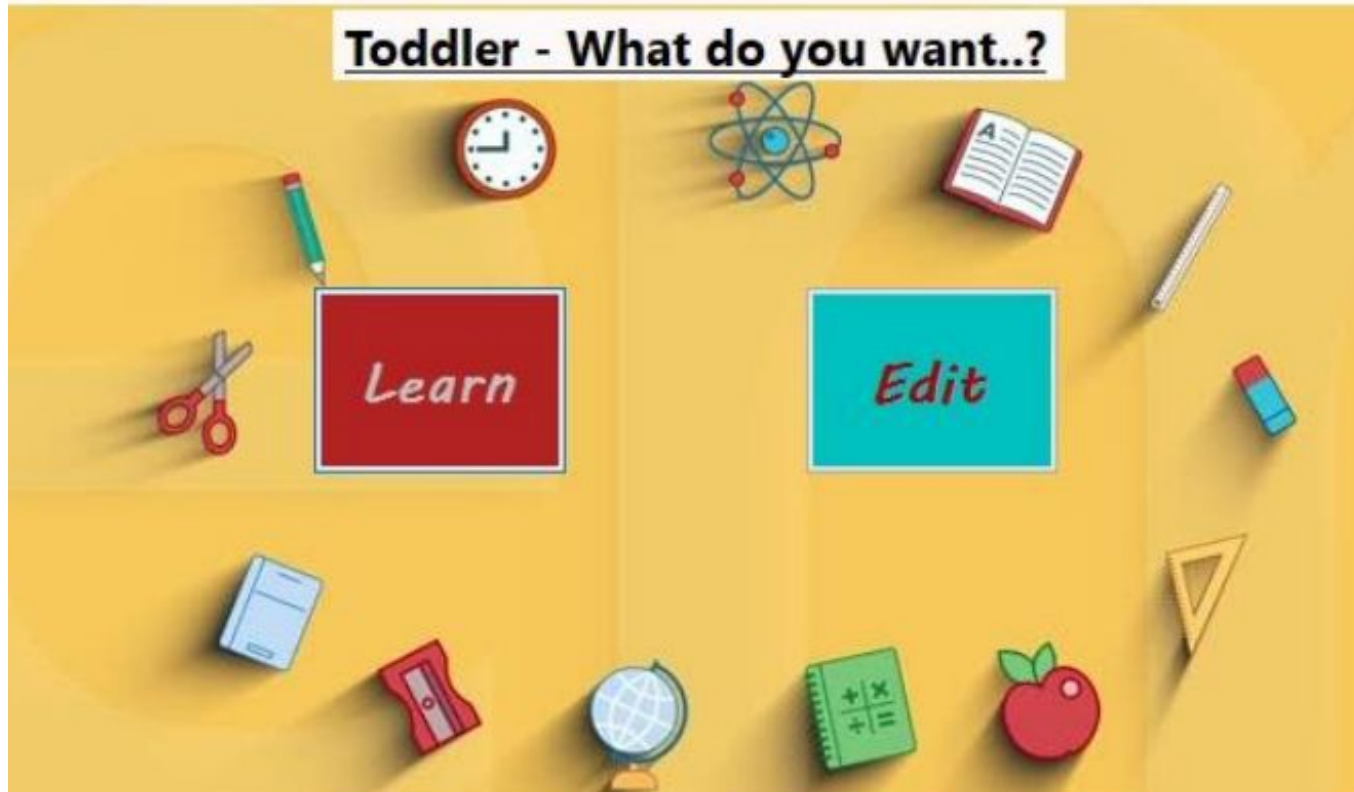
Data Management Workflow

Step-by-step administrative execution path:

LoginAuthenticate user credentials

CategorySelect target module category

CRUD Actions:



Learn

Opens the learning module for students.



Edit

Opens the protected data-management path.



Simple navigation

Two clear choices reduce user complexity.

 Student Interface Preview



Dynamic screen layout updating visual assets and metadata per selected item

 Four Primary Categories

Explore structured learning topics: Fruits, Vegetables, Animals, and Birds.

 Interactive Item Selection

Integrated list box controls allow users to seamlessly choose and inspect items.

 Visual Media Display

Detailed item information is presented alongside dedicated visual image panels.

 Category-Specific Fields

Contextual attributes automatically adjust to show nutrients, food, place, or grow type.

Admin Interface Preview



The image shows a preview of an admin interface for data management. It features a sidebar on the left with a 'Select a category' dropdown menu currently set to 'Fruits'. Below this is a list of categories: APPLE, ORANGE, and BANANA. The main area has a dark green background and contains several input fields for 'Name', 'Varieties', 'Nutrients', 'Place', 'Grow type', 'Color', and 'Photo Path'. A 'BROWSE' button is next to the 'Photo Path' field. At the bottom, there are four buttons: 'ADD NEW' (white), 'SAVE' (green), 'UPDATE' (blue), and 'DELETE' (red).

Interactive data-management interface with record selection and entry controls

Protected Access

Secure login protects the Edit workflow, preventing unauthorized administrative changes.

Item & Category Selection

Select a category and enter detailed item attributes into dedicated data fields.

Complete Record Actions

- **Add New:** Clears fields for a new record entry
- **Save & Update:** Stores new or modifies selected records
- **Delete:** Safely removes selected records

FRUIT Table

Fields: name • varieties • nutrients • place • growtype • photo • color

Category-specific: **nutrients, growtype**



VEGETABLE Table

Fields: name • varieties • nutrients • place • growtype • photo • color

Category-specific: **nutrients, growtype**



ANIMAL Table

Fields: name • varieties • special characteristic • place • food • photo • color

Category-specific: **special characteristic, food**



BIRD Table

Fields: name • varieties • special characteristic • place • food • photo • color

Category-specific: **special characteristic, food**



Database Architecture & Backend

MS Access is used as the database backend. The same CRUD pattern is applied across categories, with category-specific fields.

Dashboard Tests

STATUS: PASSED

- **Learn Form Navigation:**
Opens the Learn form reliably from the main dashboard.
- **Edit Path:**
Successfully routes to the administrative edit interface.
- **Navigation Safeguards:**
Unintended and invalid navigation attempts are verified and restricted.

Learn Tests

STATUS: PASSED

- **Category Listing:**
Each category dynamically lists all corresponding item records.
- **Detail View:**
Selecting any item correctly populates and displays detailed attributes.
- **Test Execution:**
All documented learning test cases are completed and reported as successful.

Edit Tests

STATUS: PASSED

- **Category Selection:**
Category dropdown selection functions seamlessly.
- **CRUD Operations:**
Add, Save, Update, and Delete actions are fully tested.
- **Security & Auth:**
Login authentication and password field visibility controls pass all checks.



Visual & UI Upgrades

Improve the interface with more attractive and interactive images



Content Expansion

Expand learning content to additional items such as flowers and insects



Storage & Reporting

Increase storage capacity and reporting options



Speed & Accuracy

Provide faster retrieval and more accurate work



Richer Filtering

Add richer filtering / reporting and better-designed reports



Access Control

Further strengthen user accounts and access control



Desktop Application

TODDLER is a VB.NET desktop learning application for children.



Integrated Architecture

It combines interactive learning screens with a database-backed management module.



Core Objectives

The project targets reduced manual effort, easier retrieval and a more engaging learning process.



Verified Results

The documented tests report successful results for the main dashboard, learning and edit workflows.

Thank You